Visualising Uncertainty in Train Delay Predictions: Effects on User Trust, Comprehension, and Decision Quality

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Predictions regarding the probability of train delays can assist travellers in selecting optimal connections, purchasing suitable tickets, and possible compensation claims. However, the question of how such forecasts need to be visualised in practice in order to be comprehensible, credible, and relevant for decision-making by a broad public remains unanswered. As divergent forms of visualisation are subject to differing evaluation in both research and practice we conducted a user study on forms of uncertainty visualisation. The results show that simple, intuitive visualizations, specifically hybrid representations that combine numerical probabilities with textual explanations and recognisable colour codes, enhance user understanding, trust, and decision quality.

CCS Concepts: • **Human-centered computing → Human computer interaction (HCI)**.

Additional Key Words and Phrases: delay predictions, mobile interfaces, uncertainty visualization, risk communication

1 Introduction

Train delay forecasts are increasingly integrated into journey-planning systems. These forecasts, however, are inherently uncertain: they predict probabilities rather than certainties and often rely on noisy operational data and complex models [8]. For many people, probabilistic information is difficult to interpret [9]; unclear or poorly designed visualisations may reduce the usefulness of the forecasts or even produce misleading decisions. Consequently, the way uncertainty is communicated strongly affects whether predictive systems actually support better decisions [9]. Explicitly visualising uncertainty, in contrast, can improve long-term user trust and decision quality, whereas reliance on point estimates often results in systematic decision errors [10]. However, there are only a few studies that specifically address the representation of uncertainty for non-expert target groups [10].

2 Related Work

Covello et al. [4] states that problems in risk communication are caused, among other things, by a lack of understanding of scientific methods, models and data. Although explanations have a positive effect, empirical studies indicate that the understanding of conditional probabilities by experts and non-experts is limited, regardless of the representation [6, 7, 13, 15]. A substantial body of work suggests that explanations can positively influence trust and transparency, particularly when they are visual or verbal [5, 11, 14]. Richer explanations have been associated with higher trust and improved performance [16, 17]. However, existing work mainly focuses on expert users or abstract analytical tasks, whereas little is known about uncertainty communication in everyday mobility scenarios involving non-expert users

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[10]. We therefore investigate how different uncertainty visualizations affect trust, comprehension, and decision-making in train delay prediction interfaces.

3 Expert Interviews

Two rounds of semi-structured interviews with UI/UX and HCI experts (see 1) were conducted to identify suitable visualisation concepts and evaluation criteria [2, 12]. The visualisations were selected based on their prevalence in the literature and the study hypotheses, including the traffic light system. Eight visualisation forms (stacked bar chart, box plot, gradient plot, numerical statement, 50-dot quantile plot, pie chart, textual representation, traffic light system) were selected to compare strategies for communicating uncertainty in train delay predictions.

In the first interview round, Industry experts consistently called for clarity, contextual adaptation, and simplification. They considered text indispensable and complex diagrams cognitively overwhelming under time pressure. A hybrid approach providing additional details when needed emerged as the preferred solution. Complex statistical visualizations such as box plots or gradient plots were considered unsuitable for non-expert users.

The second sprint consisted of five interviews with UI/UX experts from mostly academic backgrounds. Interviewees prioritised intuitive, action-oriented representations with minimal cognitive load and high temporal relevance, particularly in stressful situations. Numerical-textual representations were widely accepted for their clarity and reliability, although experts noted the cognitive effort required to translate probabilities into time estimates. Traffic light systems were rated most favourably due to their intuitiveness, standardisation across cultures, and robustness under stress, subject only to the requirement that context-specific legends must ensure unambiguous interpretation. Two experts suggested dynamic uncertainty visualisations: an animated timeline showing updated delay predictions and a map-based view displaying train position and travel times.

4 User Study

The selection of visualisation formats for the user study follows the experts' clear preference for simplicity, contextual clarity, and action-oriented representations. Complex statistical encodings (e.g., box plots, gradient plots, 50-dot quantile plots) were excluded due to consistently reported cognitive overload, misinterpretation risks, and limited suitability for time-critical consumer contexts. Stacked bar charts were also de-prioritized for immediate decision scenarios because of their mathematical complexity and lack of direct temporal reference. Consequently, the user study focuses on hybrid and reduced representations that were rated as most comprehensible and stress-resistant: numerical-textual combinations, traffic light systems with contextual legends and timeline-based representations inspired by expert suggestions. In addition, the expert-proposed timeline concept was incorporated in two variants to examine whether dynamic temporal contextualisation improves situational understanding without increasing cognitive burden. According to this, six different views were created. The development of the six wireframes is largely based on the insights we gained from two rounds of expert surveys. The survey design is based on procedures for online studies [1, 3]. The study aimed to address non-expert consumers who regularly make time-critical travel decisions based on delay information. The final sample consisted predominantly of university students with academic backgrounds.

Visualization A (see 1, 2) consisted of an extensive textual information screen combined with a colour-coded traffic-light dot and probability indicators. A clear majority considered arriving on time to be unlikely or very unlikely, suggesting that the traffic light signal was interpreted as a warning indicator even if the probability values themselves were not consistently understood. Similarly, less than half stated that it was clear what the prediction meant for their decision. The most critical result concerns processing speed: half of participants disagreed that the information could be understood quickly, indicating that this visualization format is poorly suited for time-critical. The perceived cognitive effort associated with this visualization was relatively high. Furthermore, more than half wished for additional explanations, indicating that the visualization did not sufficiently communicate its meaning on its own.

Visualization B (see 3, 4) combined a colour-coded probability slider, numerical probability values, textual explanation, and delay duration expressed in minutes. Participants interpreted the scenario much more consistently than in Visualization A. A total of about 92% considered a delay of more than 30 minutes unlikely or very unlikely, and about 77% believed that the train would arrive on time. With these percentages visualization B achieved the highest comprehension scores among all tested visualizations. About 92% agreed that they understood the visualization well, while 96% stated that it was clear what the prediction meant for their decision. Similarly, a high majority agreed that the information could be understood quickly, indicating excellent suitability for time-sensitive contexts. For this visualisation participants reported very low cognitive effort. Moreover, a total of about 88% considered the prediction credible, and more than half stated that they would rely on the information in everyday situations. Confidence levels were also very high with about 88% reported feeling confident about their decision.

Visualization C (see 5, 6) used a simplified traffic light representation applied directly to the departure and arrival times, with minimal textual explanation. Participants interpreted the scenario as representing a moderate risk level, which was reflected in more heterogeneous responses like with estimates regarding punctual arrival being split, with half considering punctual arrival unlikely and a third considering it likely. On the other hand did visualization C achieve very high comprehension scores and more than 90% explicitly rejected statements indicating cognitive strain. However, only about 58% agreed that the visualization conveyed a clear risk assessment, indicating that while the representation successfully triggers action, it communicates the underlying probability less precisely.

Visualization D (see 7, 8) represented the delay uncertainty using a coloured probability bar combined with numerical probability values and textual explanations. Unlike Visualization C, the bar did not directly display delay duration in minutes, requiring users to interpret the risk magnitude indirectly. Participants interpreted the risk scenario less consistently than in Visualizations B and C. This distribution indicates a less coherent perception of risk compared to the clearer patterns observed in earlier visualizations. A similar pattern emerged for punctual arrival estimates. A majority of about 58% considered arriving on time unlikely or very unlikely, while a quarter of people remained undecided and only 15.38% expected punctual arrival. Notably, the proportion of neutral responses was relatively high across trust-related items, indicating ambivalent evaluations of the visualizations usefulness.

Visualization E (see 9, 10) displayed delay predictions using textual descriptions, numerical probabilities, and colour-coded delay values, but without any graphical representation such as a bar or slider. Participants showed the most polarized risk interpretations across all visualizations. When estimating the likelihood of a delay exceeding 30 minutes, responses were almost evenly distributed across categories. Visualization E achieved the lowest trust ratings among all tested designs. Only about 61% considered the prediction credible, while 26.92% expressed scepticism.

Visualization F (see 11, 12) extended the previous design by adding a map-based representation showing the train route and coloured segments along the track. Risk perception showed considerable uncertainty. Half of the participants considered a delay of more than 30 minutes unlikely, while 30.77% remained undecided, and 19.23% considered such a delay likely. A similar pattern emerged for punctual arrival estimates: 50% considered punctual arrival unlikely, while 30.77% selected the neutral option. These results suggest that the map did not significantly improve participants' ability to estimate delay probabilities. Processing speed was also limited: only about half stated that they could grasp the information quickly, while about 26% disagreed, indicating that the map introduced additional complexity. Visualization F produced one of the highest perceived cognitive loads among all tested designs. A total of about 42% reported that they had to actively concentrate in order to understand the visualization, while about 35% described it as mentally demanding.

5 Discussion

One of the most important findings is the strong performance of Visualization B, which combined numerical probabilities, colour cues, and short textual explanations. This design achieved the highest scores across multiple evaluation dimensions, including comprehension, processing speed, trust, and perceived decision support. Participants were able to quickly grasp the meaning of the prediction and translate it into an appropriate decision. These results suggest that redundant encoding of uncertainty information—combining graphical, numerical, and textual cues can significantly improve the usability of predictive information in mobility applications. In contrast, several visualizations demonstrated how design choices can unintentionally increase ambiguity and cognitive effort. Visualization A, which contained extensive textual explanations and multiple information elements, resulted in the longest interaction times and relatively high cognitive load. This finding highlights a common challenge in uncertainty communication: providing more information does not necessarily improve understanding. The results for Visualization F also highlight the potential risks of adding visually appealing but functionally unnecessary elements. While the route map attracted attention and was perceived as visually interesting, it did not significantly improve risk perception or decision support.

The findings demonstrate that visualization design strongly affects how users interpret delay predictions and how confidently they make decisions. Simpler, action-oriented visualizations generally enabled faster interpretation and lower cognitive load, while more complex or text-heavy representations required more effort to process. At the same time, decision correctness did not necessarily depend on complete understanding of the underlying probabilities. In several cases, participants relied primarily on visual cues such as colour signals when making decisions.

6 Conclusion and future work

This study investigated how different visualization approaches influence the interpretation of uncertainty in train delay predictions within journey planning applications. Through a user study comparing six visualization designs, the results demonstrate that the representation of predictive uncertainty significantly affects risk perception, cognitive effort, trust, and travel decisions. Among the tested designs, visualizations that combined clear colour cues with numerical probabilities and concise textual explanations proved most effective. These representations enabled participants to quickly understand the predicted delay risk and make confident decisions. In contrast, designs that introduced ambiguous visual semantics, excessive textual information, or visually complex elements reduced comprehension and increased cognitive effort. The findings highlight the importance of simple, intuitive, and semantically consistent uncertainty visualizations in mobility applications for non-experts.

Future research should validate these findings with larger and more diverse participant groups. Furthermore, studies conducted in real journey-planning environments could provide insights into how uncertainty information is interpreted under realistic conditions. Additional work could also investigate dynamic uncertainty visualizations, such as timeline-based representations, as well as their effects on decision-making and long-term user trust.

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A Expert Interviews

B Wireframes

Sprint	Role	Experience
1	Design Student & Tutor	Focus on design; 3 years tutoring first-semester design courses
1	UI/UX Design Student	Academic focus on user interface design; experience through study projects; temporary freelance UI design work for an app project
1	UI/UX Freelancer	4 years self-employed; academic background in design fundamentals; focus on user-centred apps and websites for diverse user groups, including large-scale consumer applications
1	UI/UX Freelancer	Over 1 year professional experience; self-employed; completed a two-digit number of projects across various domains including e-commerce and food
2	Design Professor	10 years professorship in interactive media design; diploma in design; founder and managing director of own design firm; consultancy and UX design
2	Design Director	Over 10 years professional experience; managing director of a communication design office; focus on concept development, print and screen media design, consulting, and project management
2	HCI Professor	Professorship in Human-Computer Interaction
2	Professor for Int. Media	16 years experience; modules in electronic user interface design; professorship in interactive media
2	Design Lecturer	15 years experience; diploma in design; lecturer and internship supervisor; research associate; head of a creative media lab; focus on visual communication, graphic design, typography, print, and design thinking

Table 1. Expert profiles across both interview sprints.

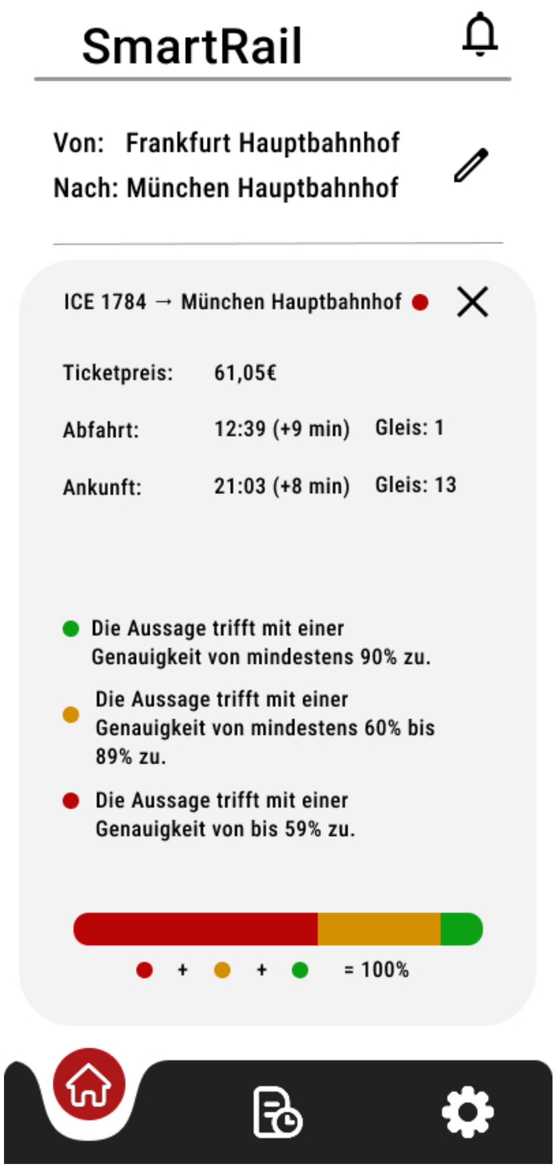

Fig. 1. Wireframe A, Details

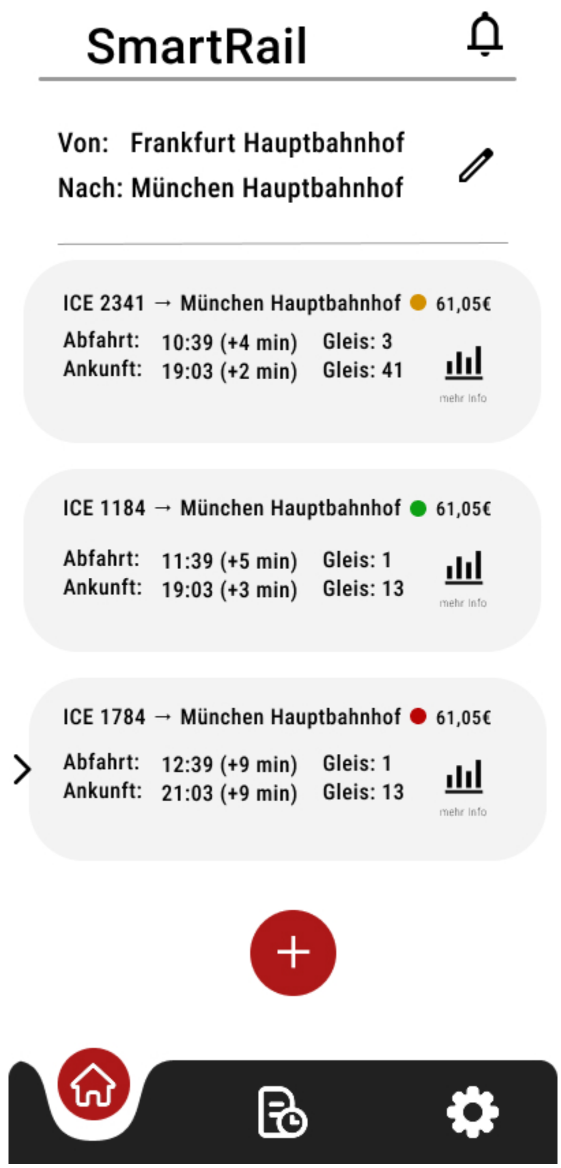

Fig. 2. Wireframe A, Overview

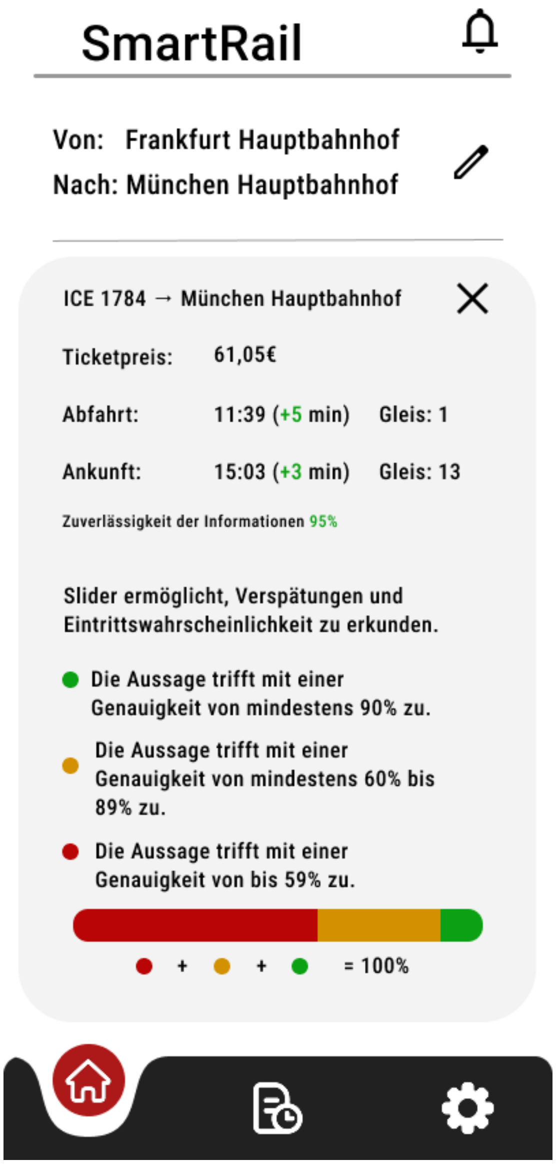

Fig. 3. Wireframe B, Details

Fig. 4. Wireframe B, Overview

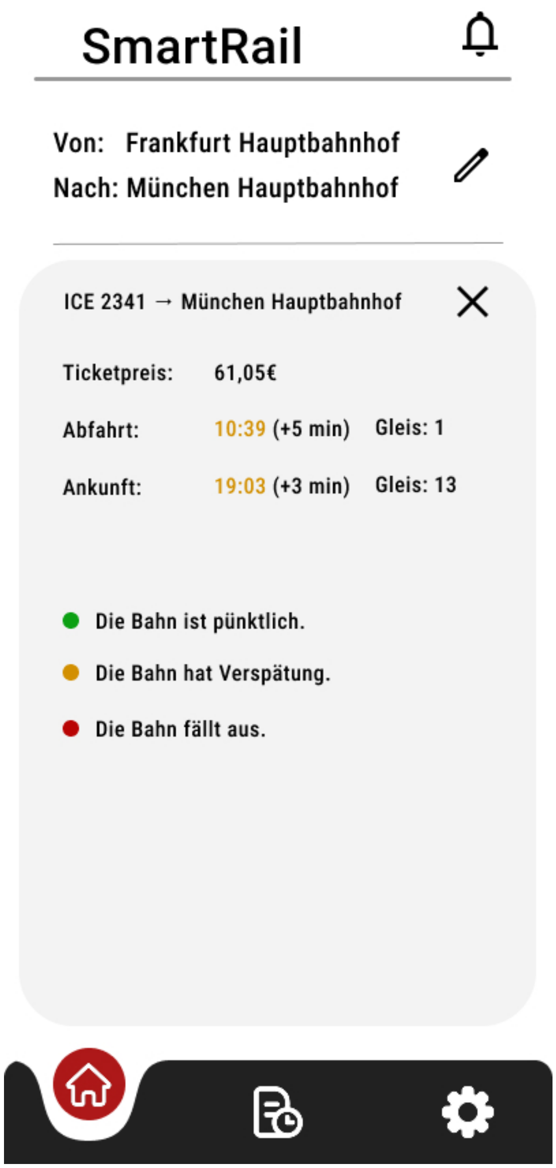

Fig. 5. Wireframe C, Details

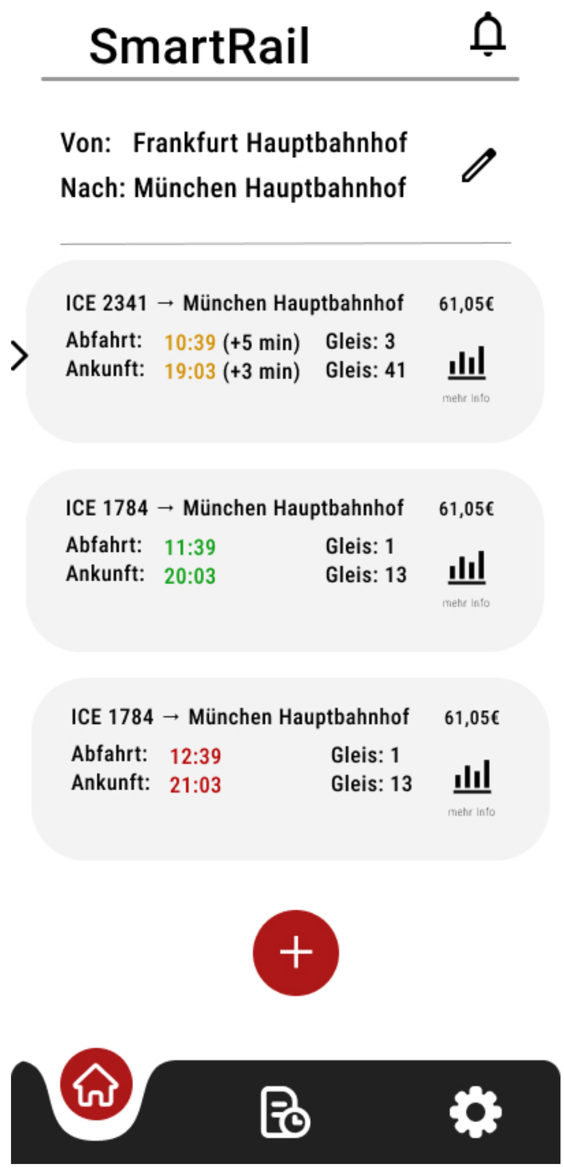

Fig. 6. Wireframe C, Overview

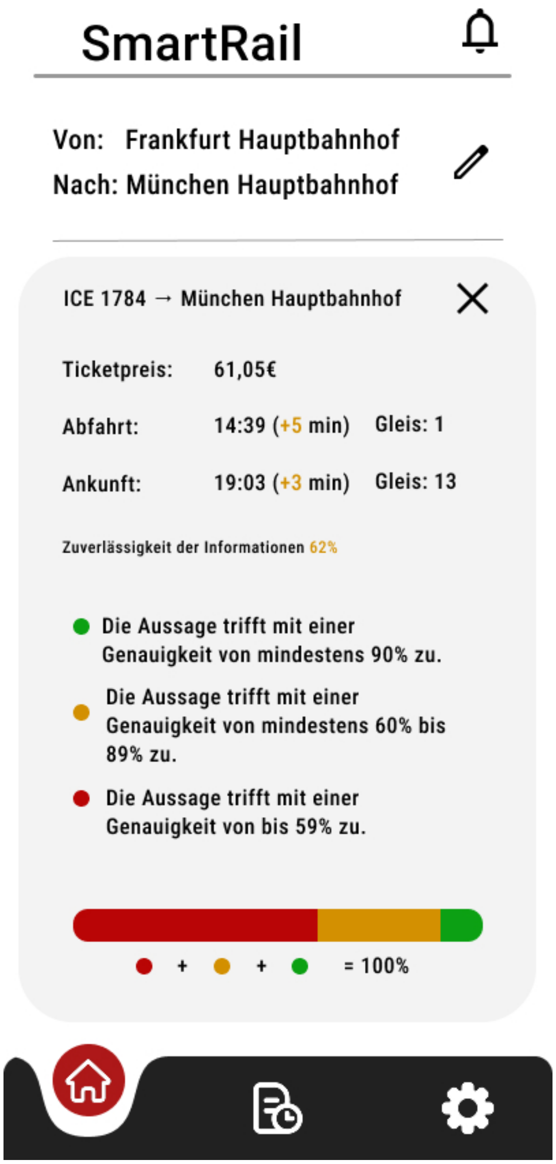

Fig. 7. Wireframe D, Details

Fig. 8. Wireframe D, Overview

Fig. 9. Wireframe E, Details

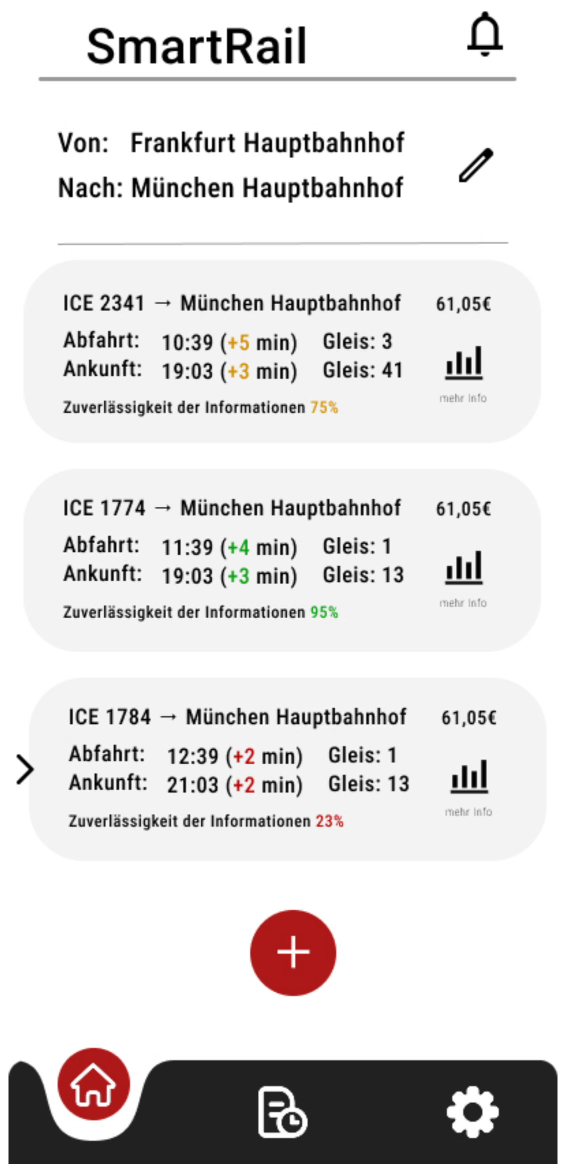

Fig. 10. Wireframe E, Overview

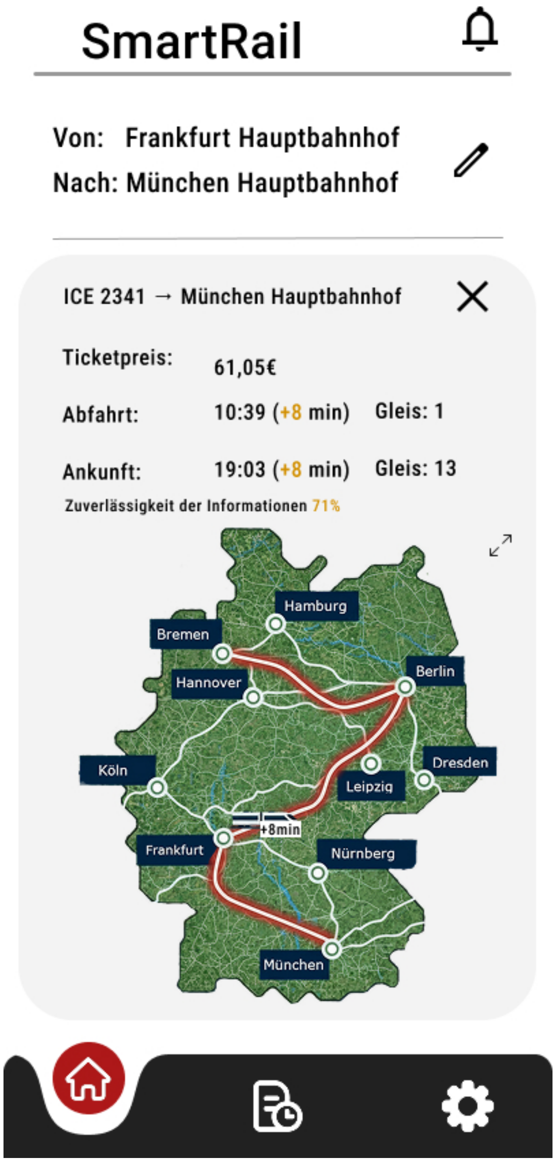

Fig. 11. Wireframe F, Details

Fig. 12. Wireframe F, Overview